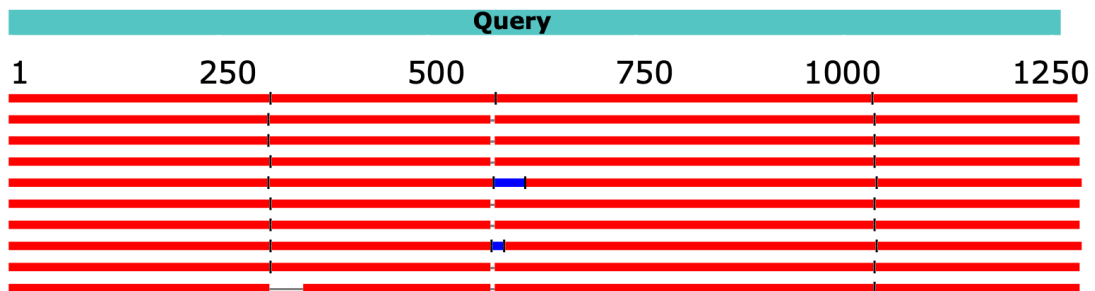


Supplementary material for Chapter 4

Figure S4.1. Five examples of chimeric inserts, illustrated using BLAST results. For each insert, the ten best BLAST hits are depicted, with the alignments of each fragment in the top hit.

Additional chimeric sequence 1:

Diploptera punctata JASPKZ010002709.1: 268362-269615 (1253 bp)



Blattabacterium sp. DPU chromosome

Sequence ID: **CP049785.1** Length: 623008 Number of Matches: 6

Range 1: 519604 to 520049

Score	Expect	Identities	Gaps	Strand	Frame
801 bits(887)	0.0()	445/446(99%)	0/446(0%)	Plus/Minus	
Query 567		GAAGAATAGGATCCATAAAATTCTAAGATTTTTGTTACAGAAGGCA		ttttttttAATTTT	626
Sbjct 520049		GAAGAATAGGATCCATAAAATTCTAAGATTTTTGTTACAGAAGGCA		TTTTTTTAAATTTT	519990
Query 627		CCTATTTTCGATAGCTTTATCAACAGCATCTTCTGCCATTTTCTATATGTAGTCCATTTT			686
Sbjct 519989		CCTATTTTCGATAGCTTTATCAACAGCATCTTCTGCCATTTTCTATATGTAGTCCATTTT			519930
Query 687		CCCCCTATAATACTAATAAGTCCAGAAGAACTAATCATAAGTTTATGAGATCTAGAAATA			746
Sbjct 519929		CCCCCTATAATACTAATAAGTCCAGAAGAACTAATCATAAGTTTATGAGATCTAGAAATA			519870
Query 747		TCTTTAGTTTTAGTAATATCAGAACAAGAATTTT		TAGGaaaaaaaaGAGGCCGTAATCCA	806
Sbjct 519869		TCTTTAGTTTTAGTAATATCAGAACAAGAATTTT		TAGGAAAAAAAAAGAGGCCGCAATCCA	519810
Query 807		GAGAATGCACCTTAATATATCAC		tttttttGTATGAAATACAAAATATTTGTTAAAAGTT	866
Sbjct 519809		GAGAATGCACCTTAATATATCACTTTTTTTTGTATGAAATACAAAATATTTGTTAAAAGTT			519750
Query 867		TGTAATATAAAATCTATTTCTTTTCTAAAGGTTTTGGTTCAAGAACACTTTTTTCCAAA			926
Sbjct 519749		TGTAATATAAAATCTATTTCTTTTCTAAAGGTTTTGGTTCAAGAACACTTTTTTCCAAA			519690
Query 927		AAAGTATCTGTAGTTCCAACATAAACATGATCACACCATGGGACACAAAATAACTCTT			986
Sbjct 519689		AAAGTATCTGTAGTTCCAACATAAACATGATCACACCATGGGACACAAAATAACTCTT			519630
Query 987		CCATCTGCAGTTTTTGAATAACTAT	1012		
Sbjct 519629		CCATCTGCAGTTTTTGAATAACTAT	519604		

Range 2: 498063 to 498366

Score	Expect	Identities	Gaps	Strand	Frame
538 bits(596)	5e-152()	303/305(99%)	1/305(0%)	Plus/Plus	
Query 1	CTTTTATATAaaaaaaTTATTGAAAAATTTCCAAATATTGAGTTTATCGCAAGTGGAGGA	60			
Sbjct 498063	CTTTTATATAAAAAAATTATTGAAAAATTTCCAAATATTGAGTTTATCGCAAGTGGAGGA	498122			
Query 61	ATTAGTAATATAGATGATGTAGATCAATTATTTAATTTAGGTTGTAGTGGAGTCATTATT	120			
Sbjct 498123	ATTAGTAATATGGATGATGTAGATCAATTATTTAATTTAGGTTGTAGTGGAGTCATTATT	498182			
Query 121	GGAAAAGCTGTATATGAAAATAAAATATCATTATCTGATCTTAAAGATTGGATAAGaaaa	180			
Sbjct 498183	GGAAAAGCTGTATATGAAAATAAAATATCATTATCTGATCTTAAAGATTGGATAAG-AAA	498241			
Query 181	aaaaaaTAATAATAAATCAATATGTTAGCTAAACGTATTATTCCCTGTTTGGACATTAAA	240			
Sbjct 498242	AAAAAATAATAATAAATCAATATGTTAGCTAAACGTATTATTCCCTGTTTGGACATTAAA	498301			
Query 241	AATGGAAGAACCGTAAAAGGAATAAATTTTAAACATTTAAAAGATGCAGGAGATCCAATA	300			
Sbjct 498302	AATGGAAGAACCGTAAAAGGAATAAATTTTAAACATTTAAAAGATGCAGGAGATCCAATA	498361			
Query 301	CAATT 305				
Sbjct 498362	CAATT 498366				

Range 3: 172184 to 172451

Score	Expect	Identities	Gaps	Strand	Frame
480 bits(531)	4e-134()	267/268(99%)	0/268(0%)	Plus/Minus	
Query 302	AATTGGAACAttttttttAATACCaaaaatacttcttgtaaaataaaaaaatatctctc	361			
Sbjct 172451	AATTGGAACATTTTTTTTAATACCAAAAATACTTCCTGTAAAATAAAAAAATTTATCTC	172392			
Query 362	aaaaaaaaatttcgttgagtgaggattaaaaaatattctattcaaaaaaCGGATTATCTTCT	421			
Sbjct 172391	AAAAAAAAATTCGTTGAGTGGATTAAAAAATATTCTATTCAAAAAACGGATTATCTTCT	172332			
Query 422	CATTAAAGGATCAAGAAATATTGCATTAGAAAAGCTTATTTCCCTTAATTTGATAGAAATT	481			
Sbjct 172331	CATTAAAGGATCAAGAAATATTGCATTAGAAAAGCTTATTTCCCTTAATTTGATAGAAATT	172272			
Query 482	CTTTTCTTGGTATATTGGATCTTAAAGATTAAATTTGTATACAATGATTTCATAAAATT	541			
Sbjct 172271	CTTTTCTTGGTATATTGGATCTTAAAGATTAAATTTGTATACAATGATTTCATAAAATT	172212			
Query 542	TTTCTTCATGAAAGAAATTACCAAAGAA 569				
Sbjct 172211	TTTCTTCATGAAAGAAATTACCAAAGAA 172184				

Range 4: 521046 to 521289

Score	Expect	Identities	Gaps	Strand	Frame
436 bits(483)	5e-121()	243/244(99%)	0/244(0%)	Plus/Plus	
Query 1010	TATTTAATATTCCAATTACCATGCTTCCAGAGGTAAAAATCTTCTAGTGAAATTTTGGTT				1069
Sbjct 521046	TATTTAATATTCCAATTACCATGCTTCCAGAGGTAAAAATCTTCTAGTGAAATTTTGGTT				521105
Query 1070	ATACAACAGGACATATTTTATCCCATAAAAATCCCTATATCTGGGATAGCTGGAGATCAAC				1129
Sbjct 521106	ATACAACAGGACATATTTTATCCCATAAAAATCCCTATATCTGGGATAGCTGGAGATCAAC				521165
Query 1130	AAGCCGCTCTTTTGGTCAGATGTGTACCAAAATGGGATGGTGAAAAATACTTATGGAA				1189
Sbjct 521166	AAGCTGCTCTTTTGGTCAGATGTGTACCAAAATGGGATGGTGAAAAATACTTATGGAA				521225
Query 1190	CAGGATGTTTTATGTTAATGAATGTAGGAAATAATCCTGTTTTTCTAGAAATAATTTAA				1249
Sbjct 521226	CAGGATGTTTTATGTTAATGAATGTAGGAAATAATCCTGTTTTTCTAGAAATAATTTAA				521285
Query 1250	TAAC 1253				
Sbjct 521286	TAAC 521289				

Range 5: 520139 to 520168

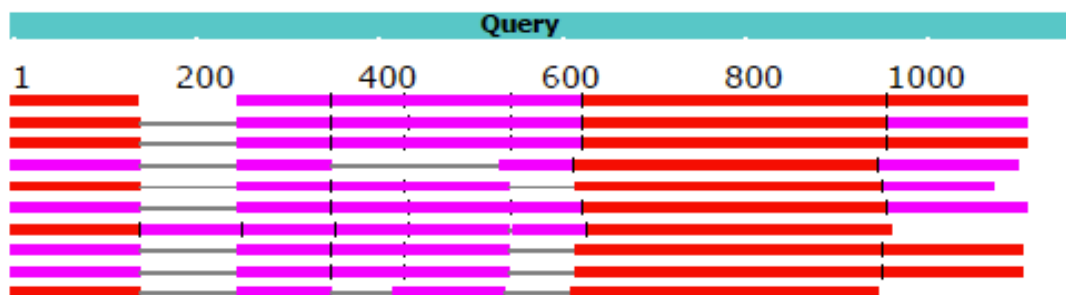
Score	Expect	Identities	Gaps	Strand	Frame
44.6 bits(48)	0.003()	29/31(94%)	1/31(3%)	Plus/Plus	
Query 460	TTTCCTTAATTTGATAGAAATTCCTTTTCTT				490
Sbjct 520139	TTTCCTTAATTTCA-AGAAATTCCTTTTCTT				520168

Range 6: 386042 to 386071

Score	Expect	Identities	Gaps	Strand	Frame
42.8 bits(46)	0.011()	28/30(93%)	1/30(3%)	Plus/Plus	
Query 530	TTTCATAAAATTTTCTTCATG-AAAGAAA				558
Sbjct 386042	TTACATAAAATTTTCTTCATGAAAAGAAA				386071

Additional chimeric sequence 2:

Neogeoscapheus dahmsi contig_7972: 1051443-1052600 (1157 bp)



Blattabacterium cuenoti strain BPAU chromosome

Sequence ID: CP142618.1 Length: 632444 Number of Matches: 6

Range 1: 340984 to 341323

Score	Expect	Identities	Gaps	Strand	Frame
457 bits(247)	9e-128()	312/341(91%)	13/341(3%)	Plus/Minus	
Query 596	CTATTAATTTTAAATT--aaaaaaaaa-TA-TTTACGCTATCAACAATGTTGTTCTATTTT	651			
Sbjct 341323	CTATTAATTTTAAATTTTAAAAAAAATTAATTCAGCTATCAACAACGTTGTTCTATTTT	341264			
Query 652	TTGATAAAATAGAATTAACAACATAATATGTTGCAATATAATTTGATATCAAATCAAAttt	711			
Sbjct 341263	TTGATAACAAAAAATTAACAACATAATATGTTGGAATATAATTTGATATCAAATCAAATTT	341204			
Query 712	ttttAT-AAAAAATGA-CATTCTCCATTATGGAGATTATACCATATGTAGTAAAGAA-Ct	768			
Sbjct 341203	TTT-ATAAAAAAAGCATTCTCCATTATGGAGATTATACCATATGTAGTAAAGAAGGT	341145			
Query 769	-ttttttACATCAATA---CAAAAATAGAGTTAAACATGAAATAAAATTGACTGGAAAA	824			
Sbjct 341144	CTTTTTACATAAATAAAAAAATAGAGTTAAACATGAAATCAAATTGATTGGAAAA	341085			
Query 825	AATTATACTGTATATGCTAATACGTTAGTATATCTATTTAAACAAGATCAAATCCATTT-	883			
Sbjct 341084	AATTATACTGTATATGCCAATACGTTAGTATATCTATTTAAACAAGATCAAATCCATTTT	341025			
Query 884	CATGATCCCGCATCATAGTACAAAATACAAATTTTGATAA	924			
Sbjct 341024	CATGATCCCGCTATCATAGTACAAAATACAAATTTTGATAA	340984			

Range 2: 586822 to 586969

Score	Expect	Identities	Gaps	Strand	Frame
213 bits(115)	2e-54()	138/149(93%)	2/149(1%)	Plus/Minus	
Query 924	ACCAAAGTTTTTTCAGAAATTAATTTGTAAGTTGATCAAAACAATTCATAATTCTGAAA	983			
Sbjct 586969	ACCAAAGTTTTTTCAGAAATTAATTTGTAAGTTGATCAAAACAATTTACAATTCATAAA	586910			
Query 984	CGTTGACCCGGTGTTATAGGCTTTAAAttttttttATTGACATTAATTAAAAATCTTCTTT	1043			
Sbjct 586909	CGTTGACCAGGTGTTATAGGTTTAACTTTTTT-ATTGACATTAGTTAAAAATCTTTTTT	586851			
Query 1044	GTTTAAAAAATCAATTTTTT-GATATTCAA	1071			
Sbjct 586850	GTTTAAAAAATCAATTTTTTGTATCTTCAA	586822			

Range 3: 330048 to 330165

Score	Expect	Identities	Gaps	Strand	Frame
169 bits(91)	5e-41()	109/118(92%)	0/118(0%)	Plus/Plus	
Query 406	TAGATTTAAAAATTCGACAATCTATAATTAACATCATGTGTCACAATATTATTTCTCCCG				465
Sbjct 330048	TAGATTTAAAAATTCGACAATCTATAATTAACATCATGTGCCACAGTATTATTTCTCCCA				330107
Query 466	CATACAATAGATTATAAAATTTACCAATTTTGTGTTGATGTAATTTGCATTTAACAC				523
Sbjct 330108	CATACAATATGTTATAAAATTTATCAACTTTCTTTATGTAATTTGCATTTAACAC				330165

Range 4: 527873 to 527969

Score	Expect	Identities	Gaps	Strand	Frame
158 bits(85)	1e-37()	93/97(96%)	0/97(0%)	Plus/Plus	
Query 241	AAAAAAGTTTTTCAGTTCTCAAACTTTATTGATTAAAAAATTGCAATCTATTTGAATGA				300
Sbjct 527873	AAAAAATTTTTTCAGTTCTCAGAACTTTATTGATTAAATAAATTGCAATCTATTTAAATGA				527932
Query 301	AAAATTATGTCATAATTCTGTTTTGAATGaaaaaaaa				337
Sbjct 527933	AAAATTATGTCATAATTCTGTTTTGAATGAAAAAAAA				527969

Range 5: 623980 to 624056

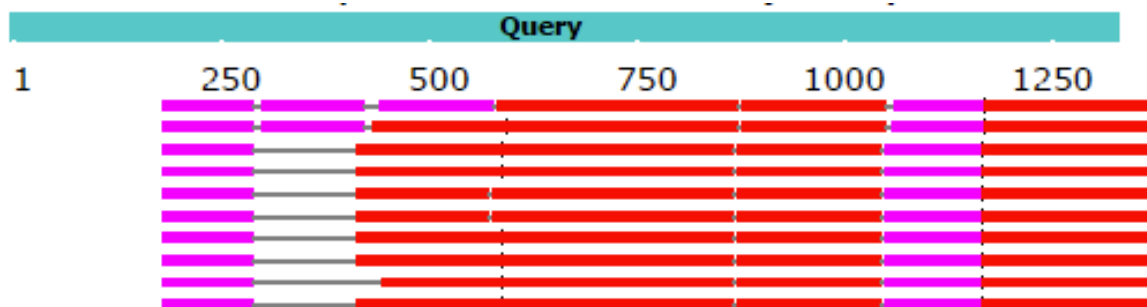
Score	Expect	Identities	Gaps	Strand	Frame
119 bits(64)	5e-26()	73/77(95%)	2/77(2%)	Plus/Plus	
Query 339	aaTCTTCG-AAAAAATTACATTTATTTTGTGTTGGATAAAATGATCAGCTAAAAATT				397
Sbjct 623980	AATCTTCGAAAAAAATACATTTATTTTGTGTTGGATAAAATGATCAGCTAAAAATT				624039
Query 398	TTAAGAATA-GATTTA				413
Sbjct 624040	TTAAGAATAAGTTTTA				624056

Range 6: 48695 to 48769

Score	Expect	Identities	Gaps	Strand	Frame
115 bits(62)	6e-25()	71/75(95%)	2/75(2%)	Plus/Minus	
Query 524	GTTGATTACGATATCTCGAGTTTTTGTAGGTATAGAAACCACGGTGTATGACT--TAATT				581
Sbjct 48769	GTTGATTACGATATCTTAAGTTTTTGTAGGTATAGAAACCACGGTGTATGACTCCTAATT				48710
Query 582	TTCTAATTCCTTTCC				596
Sbjct 48709	TTCTAATTCCTTTCC				48695

Additional chimeric sequence 3:

Panesthia cribrata ctg. 000829F: 236708-238036 (1328 bp)



Blattabacterium cuenoti BPAA DNA, complete genome

Sequence ID: AP012548.1 Length: 632490 Number of Matches: 7

Range 1: 605592 to 605875

Score	Expect	Identities	Gaps	Strand	Frame
462 bits(250)	2e-129()	273/284(96%)	2/284(0%)	Plus/Plus	
Query 570	TTCCTTTTTTAAAGGAGGAAGAAAATATAGCTaaaaaaaGACGATCTAATCCTAAAGATGT	629			
Sbjct 605592	TTCTTTTTTAAAGGAAGAAGAAAATATAGCTAAAAAAGACGATCTAATCCTAAAGATGT	605651			
Query 630	TTCTATAACATAAGGAATAGAAatttttttttattcattcaaaaattcgaattttttttt-cg-	687			
Sbjct 605652	TTCTATAACATAAGGAATATAATTTTTTTTCCATTCAAAAATTCGAATTTTTTTTCGA	605711			
Query 688	aaaaaattcatgatttttttAAATCAAAATCTCTACGAGAATGAATCCCTTCTATTTCTTG	747			
Sbjct 605712	AAAAAATTCATGATTCTTTAAATCAAAATCTCTACGAGAATGAATTCCTTCTATTTCTTG	605771			
Query 748	AAATCCAAAAGGAAAATGAAATTTCTATATCTGATTCCGCACCTTGATAATGAGACAAATG	807			
Sbjct 605772	AAATCCAAAAGGAAAATGAAATTTTATATCTGATCCCGCACCTTGATAATGAGATAAATG	605831			
Query 808	ATCATGATCACATAACTGATAATATGTTTTATCTTCTAAATTTA	851			
Sbjct 605832	ATCATGATCACATAACTGATAATATGTTTTATCTCCTAAATTTA	605875			

Range 2: 182234 to 182431

Score	Expect	Identities	Gaps	Strand	Frame
322 bits(174)	4e-87()	190/198(96%)	0/198(0%)	Plus/Minus	
Query 1131	CTGACCCATTTTCAGATAATGCATTTTCAACTAAAATTTTAATATTTTGCCATTGTAAAA	1190			
Sbjct 182431	CTGATCCATTTTCAGATAATGCATTTTCAACTAAAAATTTAATATTTTGCCATTGTAAAA	182372			
Query 1191	ATGAAATATCTTGAAATTCATCTATGAAATAATGTTTATATTGTGTGCCTATTTTTTCAT	1250			
Sbjct 182371	ATGAAATATCTTGAAATTCATCTATGAAATAATGTTTATATTGTGTGCCTATTTTTTCAT	182312			
Query 1251	ATATTTTTGGAAATGTTCTTCAACAATTCTTTCATAAAGAATTTTATTTAATTCTGCAT	1310			
Sbjct 182311	ATATTTTTGGAAATGTTCTTCAAGAATTTCTCTCGTAAAGAATTTTATTTAATTCTGCAT	182252			
Query 1311	TTAAGAttttttttttttt	1328			
Sbjct 182251	TTAAAATAATTTTTTTT	182234			

Range 3: 103094 to 103260

Score	Expect	Identities	Gaps	Strand	Frame
270 bits(146)	2e-71()	160/167(96%)	0/167(0%)	Plus/Minus	
Query 853	AAAATAGAGATATAAGATTTTATATTTCCAATGATATTATTATGTGTAAAAGGGATGTAG	912			
Sbjct 103260	AAAATAGAGATATAAGATTTTATATTTCCAATGATATTATTATGTGTAAAAGGGATGTAG	103201			
Query 913	ATAGGAGATATAGGCATTCGTAGTCCATAATGAATTTTGCTGACTAATATGAAATCTCCT	972			
Sbjct 103200	ATAGGAGATATAGGCATTCGTAAATCCATAATGAATTTTACTGACTAATATGAAATCTCCT	103141			
Query 973	ACTAATAAAGTTCTTTTCATAGAAGGAGTGGGAATCACAGAAGATTG	1019			
Sbjct 103140	ACTAATAAAGTTCTTTTCATAGAAGGAGTGGGAATCACAAAAGGTTG	103094			

Range 4: 24045 to 24160

Score	Expect	Identities	Gaps	Strand	Frame
198 bits(107)	7e-50()	114/117(97%)	1/117(0%)	Plus/Minus	
Query 296	TGAAGTTTGAATGCGTTCTTCTTGAAACGTTTTTTTTtaacgttttttGGAAAAACGAG	355			
Sbjct 24160	TGAAGTTTGAATGCGTTCTTCTTGAAACGTTTTTTT-AACGTTTTTGGAAAAACGAG	24102			
Query 356	TAATGATATACATAAAAGGAATAGTGAAAGAACTATAAAAGATAATTTTTAATGAA	412			
Sbjct 24101	TAATGATATACATAAAAGGAATAGTGAAAAAACTATAAAAGATAATTTTGATGAA	24045			

Range 5: 78195 to 78326

Score	Expect	Identities	Gaps	Strand	Frame
193 bits(104)	3e-48()	125/134(93%)	5/134(3%)	Plus/Plus	
Query 434	ATAAAAAGAAAATGAATCTA	-----TTTTTTTATCTTTTATAAAATTCATGATAGATTTGTTA			490
Sbjct 78195	ATAAAAA-AAAA-GAATCTATTT	TTTTTTTATCTTTTATAAAATTCATGATAGATTTGTTA			78252
Query 491	GAACCAAAAAAGAATTAATATATCCCCCTTTTGTAAACAGTCTCTCCTGTAACATAATCCT				550
Sbjct 78253	GAACCAAAAAAGAGTTAATATATCCCCCTTTTGTAAACAGTTCTCCTGTCACTAATCCT				78312
Query 551	ATGACTTTTCTTGT	564			
Sbjct 78313	ATGACTTTTCTTGT	78326			

Range 6: 504481 to 504582

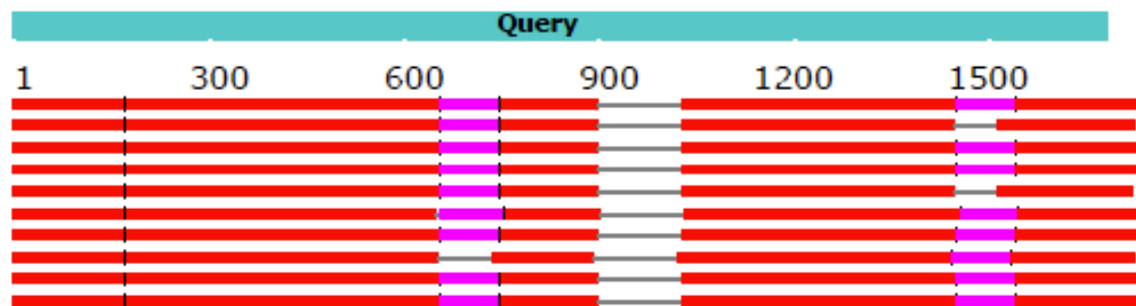
Score	Expect	Identities	Gaps	Strand	Frame
161 bits(87)	1e-38()	97/102(95%)	0/102(0%)	Plus/Plus	
Query 183	ctaaaattTATGAATGTTATACAAAACGTTGTTTTCAAGCAAACGCGTTAGATTTTGTG				242
Sbjct 504481	CTAAAAATTTATAAATTTTATACAAAACGTTGTTTTCAAGCAAATGCAATAGATTTTGTG				504540
Query 243	ATATATTACTTCATACTAATTATTAAtttttttATTTTCCAA	284			
Sbjct 504541	ATATATTACTTCATACTAATTATTTATTTTATTTTCCAA	504582			

Range 7: 174666 to 174765

Score	Expect	Identities	Gaps	Strand	Frame
158 bits(85)	1e-37()	95/100(95%)	0/100(0%)	Plus/Minus	
Query 1032	TAATAAATAATTATAAAAGTTTGTTAATAGATTGAATTAATATTTCTTTCTCTATTATAG				1091
Sbjct 174765	TAATAAATAATTATAAAAGTTTGTTAATAGATTGAATTAATATTTTCTTTCTATTATAG				174706
Query 1092	AAACTttttttGATAAAGATATTGGAGTCTCCTCTGAATC	1131			
Sbjct 174705	AAACTTTCTTTGATAAAGATACTGGAGTCTCCCTGAATC	174666			

Additional chimeric sequence 4:

Parapanesthia gigantea contig_13648: 515186-516859 (1673 bp)



Blattabacterium cuenoti strain BMK chromosome

Sequence ID: CP142624.1 Length: 632180 Number of Matches: 7

Range 1: 251744 to 252223

Score	Expect	Identities	Gaps	Strand	Frame
837 bits(453)	0.0()	472/481(98%)	2/481(0%)	Plus/Plus	
Query 157	CTCTTCGAGTATTTTCTAGATATTTTGGAAATTTTCAATAATATAACATCTACTACTACAA	216			
Sbjct 251744	CTTTTCGAGTATTTTCTAGATATTTTGGAAATTTTCAATAATATAACATCTACTACTACAA	251803			
Query 217	AAAATAGATTCCGAGTTACGGTTTCAAAAATAATGGATTTTGAACATCCAGGAATTTTGA	276			
Sbjct 251804	AAAATAGATTCCGAGTTACGGTTTCAAAAATAATGGATTTTGAACATCCAGGAATTTTGA	251863			
Query 277	ATCAAGCAATTATGGATTTAGGTTCTATTTTATGTATTCAAAAAGTCCTAAATGTTTAT	336			
Sbjct 251864	ATCAAGCAATTATGGATTTAGGTTCTATTTTATGTACTCCAAAAAGTCCTAAATGTTTAT	251923			
Query 337	TATGTCCAGTTCAAGATTCTTATTTTCTATTCAAAAATGGAAGTGTACATAAATTACCT	396			
Sbjct 251924	TATGTCCAGTTCAAGATTCTTGTTTTCTATCC-AAAATGGAAGTGTACATAAATTACCT	251982			
Query 397	GT-aaaaaaataaaaaaaTTCATAAGACATAGAttttttATTATCTTTTCATATGTGAT	455			
Sbjct 251983	GTAAAAAAATAAAAAGATTTCATAAGACATAGATTTTATTATCTTTTCATATGTGAT	252042			
Query 456	CATAACAAAAATATTTGTTTAAATAAAAGATCAACTAAAGATATATGGAAGGGTCTTTAT	515			
Sbjct 252043	CATAACAAAAATATTTGTTTAAATAAAAGATCAACTAAAGATATATGGAAGGGTCTTTAT	252102			
Query 516	GATTTTCCTTTAATAGAATCGAAACAAAATCTTTCAATTCATGAAATCATAGATGAAACT	575			
Sbjct 252103	GATTTTCCTTTAATAGAATCGAAACAAAATCTTTCAATTCATGAAATCATAGATGAAACT	252162			
Query 576	TGGGAAAAATTTAGAGTGAGGTTTTCTAAAAATGTGATTTATAAAGTAGAACAAAACTG	635			
Sbjct 252163	TGGGAAAAATTTAGAGTGAGGTTTTCTAAAAATGTGATTTATAAAGTAGAACAAAACTG	252222			
Query 636	A 636				
Sbjct 252223	A 252223				

Range 2: 372686 to 373117

Score	Expect	Identities	Gaps	Strand	Frame
542 bits(293)	4e-153()	392/434(90%)	30/434(6%)	Plus/Plus	
Query 999	at t t t t a a a a t a a t a t c c a t t t t a t g a a a a t g g t t t t t t a a a a t t a a c a g a t t t t g a a t t t				1058
Sbjct 372686	A T T T T A A A A A T A A T A T C C A T T T A T G A A A A T G G T T T T T A A A A T T A A C A G A T T T T G A A T T T				372745
Query 1059	t t a a t t t c a g a a a a a c a a a a a t A G T A T C T A T T A G T C A T A T A T A T C T A A T G T T T T A G G G A				1118
Sbjct 372746	T T A A T T T C A G A A A A A C A A A A A T A G T A T C T A T T A G T C -- A T A T A T C T A A T G T T T A G G G A				372803
Query 1119	T A A T T A A T C C T G T T C A A G A T A T T A T T A A A A A A G C T C G T G A A T A T T G A G C T T T A G T T T T G A				1178
Sbjct 372804	T T A T T A A T C C T G T T C A A G A T A T T A T T A A A A A A G C T C A T G A A T A T G G A G C T T A G T T T T G A				372863
Query 1179	T T G A T G G A G T T C A A G T C C C T T C T A A T T T A G A C T T A G -----				1214
Sbjct 372864	T T G A T G G A G C T C A A G T C C C T T C T A A T T A G A C T T A G A T G T A C A A A A T T T A A A T G T T G A T T				372923
Query 1215	--- A T G T T T T T T C T G C T C A T A A A A T G T A T G G A C C T A C T G G A A T T G G T A T A T T A T A T G T a a				1271
Sbjct 372924	T T T A T G T T T T T C T G C T C A T A A A A T G T A T G G A C C T A C T G G A A T T G G T A T A T T A T A T G G A A				372983
Query 1272	a a a a a a a a a - a a T A G A A A A A T T A T A T C C T T A T C A A T T T G G A G G G G A A A T G A T T A A A A A T G				1330
Sbjct 372984	A A A A A A A A A T A T T A G A A A A A T T A T A T C C T T A T C A A T T T G G A G G G G A A A T G A T T A A A A A T G				373043
Query 1331	T G A G T T T T G A T A A A G C A A C T T A C T C A G A T A T A T C G T T T A A T T T T T A G G C A G G A A C T C C A A				1390
Sbjct 373044	T G A G T T T T G A T A A A A C A A C T T A C T C A G A T A T A C C G T T T A A A T T T G A G G C A G G A A C T C C G A				373103
Query 1391	A C A T A G A A G G A A T T 1404				
Sbjct 373104	A T A T A G A A G G A A T T 373117				

Range 3: 306075 to 306279

Score	Expect	Identities	Gaps	Strand	Frame
337 bits(182)	2e-91()	198/205(97%)	3/205(1%)	Plus/Plus	
Query 1472	T T G T G T T T C T A C T T T T T T A A A A A --- C T T C A T T T T T A T A T A C T A T A G G A A A A T T T T T C C A				1528
Sbjct 306075	T T G T G T T T C T A C T T T T T T A A A A A T T T C T T C A T T T T T A T A T A C T A T A G G A A A A T T T T T C C A				306134
Query 1529	T T C T T C T T C T A A A A A A G A A C T C A A T A C G T T C C A T T T A A T A T T A C T T G C C T C A T T A T A T T T				1588
Sbjct 306135	T T C T T C T T C T A A A A A A G A A C T C A A T A T G T T C C A T T T A A T A T T A C T T G C C T C A T T A T A T T T				306194
Query 1589	T A C A T G A A G A A T A T T T T A A t t t t t t t t t t t t c c a t a t t t t t t t a c a t c a t c a t t a c t a a t a a t				1648
Sbjct 306195	T A C A T G A A G A A T A T T T T C A T A T T T T T T T C C A T A T T T T T A C A T C A T C A T T A C T A A T A A T				306254
Query 1649	t t c t t c t t t t t t c c a a a t t t t t t t c T G 1673				
Sbjct 306255	T T C T T C T T T T T C C A A T T T T T T C T G 306279				

Range 4: 596310 to 596473

Score	Expect	Identities	Gaps	Strand	Frame
259 bits(140)	4e-68()	157/165(95%)	2/165(1%)	Plus/Minus	
Query 1	AACGGCTTTT	CACATGCTTGCAATAATAATTCTCCGATTATTAtttttttt	CTTGAAGCC	60	
Sbjct 596473	AACGACTTTT	CACATGCTTGCAATAATAATTCTCCGTTATTCCTTTTTTT	-TTGAAGCC	596415	
Query 61	TCAAGAAATAAACTTTT	CAAAATTGTCATTCCAATATTCCATAAGT-TATCTTGCTTTTTGT	119		
Sbjct 596414	TCAAGAAATAAACTTTT	CAAAATTGTCGTTCCAATATTCCATAAGTATATCTTGCTTTTTGT	596355		
Query 120	TTTTCTATCAATTGAATAAAAATATTCTGAACGTTTTCTCTTCGA	164			
Sbjct 596354	TTTTCTATCAATTGAATAAAAATATTCTGAACGTTTTCTCTACGA	596310			

Range 5: 181204 to 181348

Score	Expect	Identities	Gaps	Strand	Frame
235 bits(127)	7e-61()	139/145(96%)	0/145(0%)	Plus/Minus	
Query 722	TGTATTTTCATTTT	AGTAGACCTGATCCTCCATGAAAACTAAAGACACTGGttttttt	781		
Sbjct 181348	TGTATTTTCATTTT	AGTAGAACCTGATCCTCCATGAAAACTAAAGACACTGGTTTTTTT	181289		
Query 782	GTCTTAGTATGAAATTTGTTTTGTATATATTCTTGTGTATTTTTTAAAAATTCGGGATGA	841			
Sbjct 181288	GTCTTAGTATGAAATTTT	TGTATATATTCTTGTGTATTTTTTAAAAATTCGGACGA	181229		
Query 842	AGCATAACATTTCCAGGTTT	tataaa	866		
Sbjct 181228	AGCATCACATTACCAGGTTT	TATAAA	181204		

Range 6: 160006 to 160092

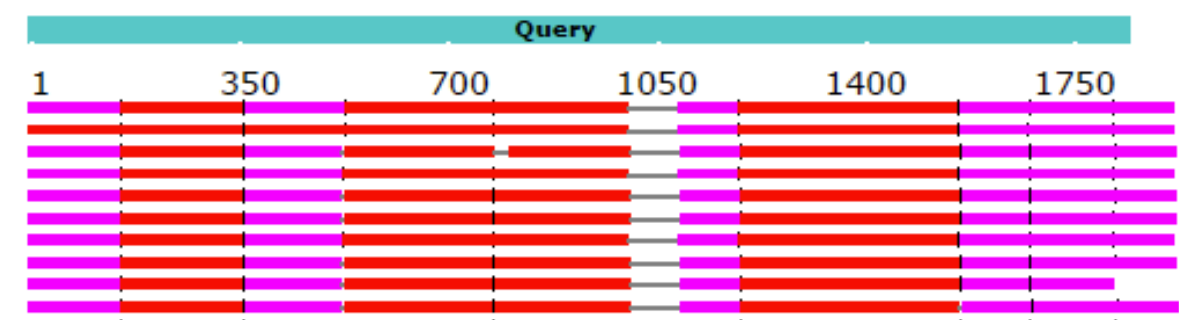
Score	Expect	Identities	Gaps	Strand	Frame
139 bits(75)	6e-32()	83/87(95%)	0/87(0%)	Plus/Minus	
Query 637	ATATTCAAAATCCAATTCAGGACAAGATTCTAGTTTATATAAAACAGATATGGATTTTCT	696			
Sbjct 160092	ATATTGAAAATCCAATTCAGGACAAGATTCTAGTTTATATAAAACAGATATGGATTTTTT	160033			
Query 697	TACTATTTTTCTGTAGATTGAAAATG	723			
Sbjct 160032	TACTATTTCTTCTATAGATTGAAAATG	160006			

Range 7: 350656 to 350738

Score	Expect	Identities	Gaps	Strand	Frame
106 bits(57)	6e-22()	76/85(89%)	2/85(2%)	Plus/Plus	
Query 1405	GTTTTTATTCAAGATCCAAAACCTGGAT	taaaaaaaTTAAAAATTTTTGGGTAAATATGT	1464		
Sbjct 350656	GTTTTTATTCAAGATCCAAAACCTGGAT	AAAAAAATATAAAA-TTTTGGATAAATATGT	350714		
Query 1465	AGAAATCTTGTGTTTCTACTTTTTT	1489			
Sbjct 350715	AGAAATCTTTATA-CTACTTGT	350738			

Additional chimeric sequence 5:

Panesthia lata contig_1558: 4419266-4421106 (1840 bp)



Blattabacterium cuenoti BPAY DNA, complete genome, strain: BPAY
Sequence ID: AP014609.1 Length: 632370 Number of Matches: 10
Range 1: 51691 to 52053

Score	Expect	Identities	Gaps	Strand	Frame
543 bits(601)	9e-149()	338/363(93%)	0/363(0%)	Plus/Plus	

Query	1141	CTCAAGTAATAGGAATTTGGAACCTTGTCTATAGAATTTATACGAAAATCAGATGAAA	1200
Sbjct	51691	CTCAAGTAATAGAAAATTTGGAATCTTGTCTATAGAATTTATACGAAAATCAGATGGAA	51750
Query	1201	CATTGGAAAACTTTATACAAAACATTTAGATACAGGAATGGGTTTGAAAAGATTATGCA	1260
Sbjct	51751	CATTGGAAAACTTGACACAAAACATGTAGATACAGGAATGGGATTGAAAAGATTATGCA	51810
Query	1261	TGGTTTTACAAGGAAAAATATCTAGTTATGAACTGATAATTTTTATCCTATTATTCAAT	1320
Sbjct	51811	TGGTATTACAAGAAAAAATTTCTAGTTATGAACTGATATTTTTATCCTATTATTCAAG	51870
Query	1321	ATATAGAAGACGATATAGGAAATGTTTATAATATAAAGGATTTTAAATCAACATGTATCTA	1380
Sbjct	51871	ATATAGAATACGATTTAGGAAATGTTTATAATAGAAAAGATTTTAAATCAAAATGTATCTA	51930
Query	1381	TACGGATTATTGCAGATCATCTAAGAGCTATTGTTATTTCTATTTTAGATGGACAATTAC	1440
Sbjct	51931	TACGGATTATAGCAGATCATCTAAGAGCTATTGTTTTCTATTTTAGATGGACAATTAC	51990
Query	1441	CATCCAATAACGGAGCTGGTTATGCGATAAGAAGGATACCTAGAAGAGCCCTTATTTACT	1500
Sbjct	51991	CATCCAATAACGGAGCTGGTTATGTGATAAGAAGAATACCTAGAAGAGCCGTATTTTCT	52050
Query	1501	GTA 1503	
Sbjct	52051	GTA 52053	

Range 2: 407510 to 407734

Score	Expect	Identities	Gaps	Strand	Frame
288 bits(319)	3e-720	201/225(89%)	4/225(1%)	Plus/Plus	
Query	743	aaaaatcttaacaatttaaaaaaaaaT--AGtttttgagcatttaatttttaattttct	800		
Sbjct	407510	AAAGATTTTATAATTTTAAAAAAATTTAGTTTTGGAGCATTTAATTTAAGTTTCT	407569		
Query	801	ataattaaaattgtattatataactaattttgttctataagaaattttctgactatattt	860		
Sbjct	407570	ATAATTAAAAATTTATTACGTACTAATTTTGTCTATGAGAAATTTCTGACTATATTT	407629		
Query	861	tttgtagctttatttaattttattaaatattttCTTGGTTAAGGACCAAAG-CTCTTCTC	919		
Sbjct	407630	TTAGTACGTTTGTTTAATTTTATAAATACCTTCTGGTTAGGACCAAAACTCTTCCC	407689		
Query	920	CCTCGTCTAAAAATA-GAATTCTTATGTCCCTTTTCTGGAGCCT	963		
Sbjct	407690	CCTCCTCTAAAAATAGGATTTTTATGTTCCCTTTCTAGAGCCT	407734		

Range 3: 449834 to 450074

Score	Expect	Identities	Gaps	Strand	Frame
283 bits(313)	1e-700	214/247(87%)	14/247(5%)	Plus/Plus	
Query	505	ataaaatataaaatattccctttatgaaaaatttttGCTTTGGCAAACAGAATTCCTTC	564		
Sbjct	449834	ATAAAATGTAATAACTTCCCTTATGAAAAATTTT-GCTTTGGCAAACAGAAATTCCTTT	449892		
Query	565	T-TTATGTTTATGGATCGAATATGATTTGTAGAAATTTCACTACTAAAT-CATTAA----	618		
Sbjct	449893	TCTTATTTTATGGATCGAATATGATTTGTAGAAATTTCAATGCTAAAAACATTAATAAT	449952		
Query	619	-ttttttGTTCAATTTTAAATATATAACTTATAaaaaaaaTAGAAATTTCAATACCTTCA	677		
Sbjct	449953	GTTTTTTTTTCAATTTATAATAT-----TTATAAAAGAAATAGAACTTCAACACTTCA	450007		
Query	678	GCTAAAATTATAATTGCTCCTCCATGtaaaaaagtgaata-gtttaaatattttagaattt	736		
Sbjct	450008	GCTAAAATTATAGTTGCTCCTCCATGTAAAAAACCAAAAGGTTGAAATATTTAGAAATTT	450067		
Query	737	ataggca 743			
Sbjct	450068	ATAGGCA 450074			

Range 4: 539533 to 539742

Score	Expect	Identities	Gaps	Strand	Frame
270 bits(299)	8e-67()	185/210(88%)	9/210(4%)	Plus/Minus	
Query 143	TTGTGGTTTTATAGCTTTATGAGGTGTAATCACAAATTTCCAAAAATTCATTTTCTTTGAT	202			
Sbjct 539742	TTGTGTTTTATAGCTTTATGAGCTTTAATCACAGTTTCCAAAAATTCATTTTCTTTGAT	539683			
Query 203	TTCTTAATTTCTCCTTATATCATACTAATAGAA-----GCTCCCAGTATAAAATC	253			
Sbjct 539682	TTCTTCATTTCTCCTTCTATCATAATAATAGAAATGATTTGAAGCTCCGACTATTAAATC	539623			
Query 254	TATATCCGCTTTCTTTAACTGGTCTAAACTAGGATTAATAATGAATTTTCCATTAAACG	313			
Sbjct 539622	TATATCCGATTTTAACTGATCTAAACTAGGATTAATAATGAATTTTCCATTAAACG	539563			
Query 314	GATAATATGTATTTCTGACATAGGAGCATT	343			
Sbjct 539562	GATAATACGTATTTCTGATATAGGACCATT	539533			

Range 5: 478954 to 479095

Score	Expect	Identities	Gaps	Strand	Frame
194 bits(214)	5e-44()	128/142(90%)	0/142(0%)	Plus/Minus	
Query 1	ATATCATAAACAttttttaattttgtttttATAACAGTAAGCATTACATTATGATTACAT	60			
Sbjct 479095	ATATCATAAATAATTTTATGTTTATATAAAGTACGCATTACATTATGATTGCAT	479036			
Query 61	AATTCTTGACTATTTGTTGGCGTGATTTTATGTGAAGTAAATTTTGTAAAGGAAAAGGA	120			
Sbjct 479035	AATTCTTGACTATTTGATGACGTGTTTATATAAAATAAATTTTAAAGGAAAAAGA	478976			
Query 121	ATCCAATTTTTTAATAATCTTC	142			
Sbjct 478975	ATCCAATTTTTTAATAACCTTC	478954			

Range 6: 575018 to 575196

Score	Expect	Identities	Gaps	Strand	Frame
183 bits(202)	1e-40()	150/179(84%)	14/179(7%)	Plus/Plus	
Query 342	TTTACAATAGAATAATTTTTGAAACTATTTCTGGACTAAAAATTTAGATCACCATTCT	400			
Sbjct 575018	TTTACGATAGAAAAATTTTTGAAACTATTTCTGGACTAAAAATTTAGATCATCATTC	575077			
Query 401	--ACTGcttta-----ttttccttttttaaaaaagatgattt-----atagaaaatttcac	449			
Sbjct 575078	ATACTGCTTTACTTTTTATTTTTAAAAAAGATGATTTATGGAAAAAAATATCAC	575137			
Query 450	tttttttttAATT-GGGGACTAAAATATT-AAtttttttcagaatatagattttattat	506			
Sbjct 575138	GATTTTTTAAATGGGGGCTAAAATATTAAATGATTTTCAAGATATAGATTTAATAT	575196			

Range 7: 229715 to 229830

Score	Expect	Identities	Gaps	Strand	Frame
170 bits(188)	6e-37()	108/116(93%)	1/116(0%)	Plus/Plus	
Query 1726	TTGCATCTTTTTAAAACTTATATTCTCCAGACTGATAAAATTGATTATAATATCCCAT	1785			
Sbjct 229715	TTGCATCTTTTTAAAAATTTATCTTCTCCAACTGATAAAATTGATTATAATATCCCAT	229774			
Query 1786	AGAATAATAGGAAC-TTTAtttttttCTTAATGTTTTCGATTTTATTAATAATAA	1840			
Sbjct 229775	AGAATAATAGGAAC-TTTATTTTCTTAATTTTCAATTTGATTAAATAATAA	229830			

Range 8: 550189 to 550326

Score	Expect	Identities	Gaps	Strand	Frame
165 bits(182)	3e-350	123/141(87%)	5/141(3%)	Plus/Plus	
Query 1605	CAGtttttccttttttttaaaaaaaaaaTTTAAATTTAGAAATCAA-ATCAATTTGTTGT	1663			
Sbjct 550189	CAGTTTTTCTTTTTTT---AAAGAAAAATTTAAATTTTGAATCAATATAAATTTGTTGT	550245			
Query 1664	AAGTAAATGAGTCATCC-GAAATAAAAATTTCAATTTCAAATGAAAAATAGATTTATATG	1722			
Sbjct 550246	AAGCAAATGATACATCCCGAAATAAAAATTTCAATTTCAAATGAAAAATAGATTTATATC	550305			
Query 1723	TAATTGCATCTTTTTAAAAAC	1743			
Sbjct 550306	TCATTTGTTTCATTTGAAACAC	550326			

Range 9: 200638 to 200745

Score	Expect	Identities	Gaps	Strand	Frame
149 bits(164)	2e-300	99/109(91%)	1/109(0%)	Plus/Minus	
Query 1503	AAGAAATGGGAATCAGGAGGTCCTTTTATTGGAAATACCTTGGGaaaaaaaTGAATCAATG	1562			
Sbjct 200745	AAAAAATGGGAATTAGGAGATCCTGTTATTCGAAACACTTGG-AAAAAATGAATCAATG	200687			
Query 1563	GGTTTATAATGGGTTTGAAGAACTTATAAAAAATTTGGGAATCAGtttt	1611			
Sbjct 200686	GGTTTATAATGGATTGAAGAACTTATAAAAAATTTGGGAATTAGTTT	200638			

Range 10: 165748 to 165849

Score	Expect	Identities	Gaps	Strand	Frame
113 bits(124)	1e-190	86/102(84%)	10/102(9%)	Plus/Minus	
Query 1049	aacacattttgaaatttgaatttattagatgattttttt---AATCTTTT-----TGT	1098			
Sbjct 165849	AATACATTTTAAATTTGAATTTATTAGATAATTTTTTCTAAATCTTTAATTTTGT	165790			
Query 1099	AAATCTCCGTTAAATAATTCTTTGAAACTATCTGCTGAGCCT	1140			
Sbjct 165789	AAATCTCCATTAAATAATTCTTTAAACTATCTGCTGAACCT	165748			